

ILE BICQUETTE, QC, Canada

WMO: 713850

Lat: 48.415N

Lon: 68.893W

Elev: 23

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -1.4	(d) -14.4	(e) 2.5	(f) -5.1	(g) -10.1	(h) 3.2	(i) -1.2	(j) 46.2	(k) 28.7	(l) 41.3	(m) 19.6	(n) 20.1	(o) 290	(p) 0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	66.5	59.7	64.0	58.2	61.9	57.1	61.3	64.5	59.5	62.5	57.9	60.8	10.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	77.0	62.8	58.0	71.9	61.0	56.4	67.8	59.5	27.3	64.7	26.1	62.5	25.0	61.0	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 36.9	(b) 33.8	(c) 30.2	(e) -10.0	(f) 74.2	(g) 4.1	(h) 2.8	(i) -12.9	(j) 76.2	(k) -15.3	(l) 77.8	(m) -17.6	(n) 79.4	(o) -20.6	(p) 81.4		
(4) 36.9	(5) 33.8	(6) 30.2	(7) -10.4	(8) 66.3	(9) 2.7	(10) 1.4	(11) -12.3	(12) 67.3	(13) -13.9	(14) 68.2	(15) -15.4	(16) 69.0	(17) -17.4	(18) 70.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	15.7	18.0	26.5	36.6	45.1	52.1	55.9	55.1	50.1	42.7	34.4	23.1		
	DBStd	15.20	10.19	9.30	7.91	5.01	4.26	4.13	3.66	3.53	3.87	4.32	6.13	8.53		
	HDD50	4862	1064	897	730	403	162	25	2	4	45	229	468	833		
	HDD65	9842	1529	1317	1195	853	617	386	283	308	447	691	918	1298		
	CDD50	497	0	0	0	0	11	89	185	161	48	3	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	15.0	17.7	16.9	16.1	14.3	12.2	12.7	13.7	13.4	13.9	14.9	16.5	17.3		
	PrecAvg	41.6	3.0	2.8	3.0	2.9	3.5	3.6	4.2	3.6	3.9	4.0	3.4	3.9		
(15) Precipitation	PrecMax	47.3	5.4	4.6	5.5	5.3	5.9	6.9	8.6	6.6	10.5	6.6	6.7	5.8		
	PrecMin	32.2	1.0	1.3	1.3	1.2	1.5	0.8	1.3	1.4	1.4	1.2	1.3	1.9		
	PrecStd	3.6	1.1	0.9	1.2	1.2	1.1	1.5	1.5	1.3	1.7	1.4	1.1	0.9		
	PrecStd	3.6	1.1	0.9	1.2	1.2	1.1	1.5	1.5	1.3	1.7	1.4	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	39.3	44.9	52.1	62.4	68.3	70.8	69.3	65.2	57.4	52.2	41.3		
		MCWB	39.3	37.6	40.5	44.6	53.2	57.4	63.0	60.8	58.9	53.5	48.1	39.3		
	2%	DB	36.0	35.1	40.5	47.5	57.6	64.0	66.7	65.7	61.0	53.5	46.1	37.7		
		MCWB	34.4	32.9	36.5	41.3	50.2	55.8	60.4	59.8	56.5	49.9	43.3	35.9		
	5%	DB	33.1	32.7	37.6	44.8	54.3	61.2	64.2	63.1	58.2	50.8	43.4	35.9		
		MCWB	31.7	31.1	34.6	39.7	47.9	54.6	58.9	58.4	54.4	47.6	40.9	34.0		
	10%	DB	29.9	30.7	35.6	42.5	51.7	58.8	61.9	61.1	56.1	48.4	41.3	33.9		
		MCWB	28.2	29.3	33.1	38.4	46.6	53.2	57.7	57.1	52.8	45.3	39.1	32.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	38.5	39.2	45.9	55.0	60.9	64.7	63.4	60.6	54.0	49.7	39.5		
		MCDB	41.5	39.8	41.1	50.1	58.1	65.7	67.9	66.5	63.5	55.9	52.5	40.6		
	2%	WB	34.8	33.5	36.6	42.1	51.0	57.6	61.9	60.8	57.4	50.4	43.8	36.0		
		MCDB	36.0	34.6	38.6	45.6	55.1	61.3	65.2	63.8	60.0	52.8	45.3	37.0		
	5%	WB	31.8	31.6	34.7	40.4	48.9	55.6	60.0	59.1	55.2	47.9	41.1	34.2		
		MCDB	33.1	32.8	36.6	43.4	52.9	59.3	62.9	62.1	57.6	50.2	42.6	35.2		
	10%	WB	28.0	29.5	33.0	38.8	46.9	53.9	58.3	57.5	53.3	45.7	39.3	32.2		
		MCDB	29.6	30.8	34.9	41.6	50.7	57.5	61.1	60.3	55.9	47.9	40.8	33.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.3	9.4	8.9	10.4	11.9	12.5	12.0	10.0	7.2	6.8	8.3		
		MCDBR	14.6	10.3	9.6	12.3	15.3	15.7	15.6	15.0	13.4	11.1	9.8	9.7		
		MCWBR	15.1	10.2	8.0	8.5	9.5	10.0	10.4	9.9	10.1	8.8	8.9	9.8		
	5% WB	MCDBR	15.0	10.3	9.4	11.1	14.0	14.4	14.6	13.8	13.2	10.7	9.0	9.3		
		MCWBR	15.0	10.1	8.4	8.2	9.9	10.8	11.2	10.4	10.8	9.2	8.4	9.8		
(40) Clear-Sky Solar Irradiance	taub	0.265	0.275	0.307	0.351	0.365	0.385	0.409	0.391	0.365	0.329	0.297	0.269			
	taud	2.356	2.318	2.292	2.315	2.354	2.362	2.326	2.381	2.438	2.520	2.469	2.394			
	Ebn at Noon	259	282	288	285	284	278	269	269	266	260	245	241			
	Edh at Noon	22	28	34	37	37	37	38	35	30	23	20	18			
(44) All-Sky Solar Radiation	RadAvg	401	693	1056	1392	1636	1744	1724	1476	1136	666	360	270			
	RadStd	41	77	78	120	154	127	168	97	75	53	36	24			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page